To find following the machine learning regression method using in r2 value

1.MULTIPLE LINEAR REGRESSION (R^2 VALUE) =0.935868

2.SUPPORT VECTOR MACHINE

S.NO	HYPER PARAMETER	LINEAR (R value)	RBF (R value)	POLY (R value)	SIGMOID (R value)
1	C=10	-0.039645	-0.056808	-0.053667	-0.054719
2	C=100	0.106468	-0.050726	-0.019802	-0.030454
3	C=1000	0.780284	0.006768	0.266164	0.185069

The SVM Regression **use R^2 value (nonlinear (RBF) and hyper parameter C (1000) = .78028**

3. DECISION TREE:

S.NO	CRITERION	SPLITTER	R VALUE
1	squared error	best	0.893707
2	squared error	random	0.860284
3	friedman_mse	best	0.923850
4	friedman_mse	random	0.901559
5	Absolute_error	best	0.935619
6	Absolute_error	random	0.541645
7	poisson	best	0.920397
8	poisson	random	0.901607

The Decision Tree use R^2 value (Criterion = "Absolute error", Splitter = best) = 0.935619.

4.Random Forest

S.NO	**CRITERION**	**MAX_FEATURES**	**N_ESTIMATORS**	**R_VALUE**
1	*squared_error*	Sqrt	1000	0.796700
	squared_error	Sqrt	100	0.759150
	squared_error	Log2	1000	0.796700
2	*squared_error*	Log2	100	0.759150
3	Friedman_Mse	Sqrt	1000	0.796316
4	Friedman_Mse	Sqrt	100	0.760859
5	Friedman_Mse	Log2	1000	0.796316
6	Friedman_Mse	Log2	100	0.760859
7	*absolute_error*	Sqrt	1000	0.815091
8	*absolute_error*	Sqrt	100	0.785748
9	*absolute_error*	Log2	1000	0.815091
10	*absolute_error*	Log2	100	0.785748
11	*poisson*	Sqrt	1000	0.781925
12	*poisson*	Sqrt	100	0.771764
13	*poisson*	Log2	1000	0.781925
14	*poisson*	Log2	100	0.771764

The random Forest Regression R2 value (criterion='absolute error', max_features = sqrt &log2, c=1000) **= 0.815091**

The final machine learning best method of Regression:

The Decision Tree use R^2 value (Criterion = "Absolute error", Splitter = best) **= 0.935619.**